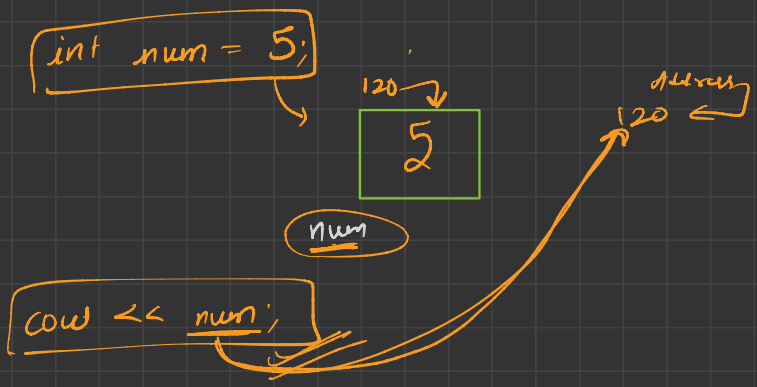
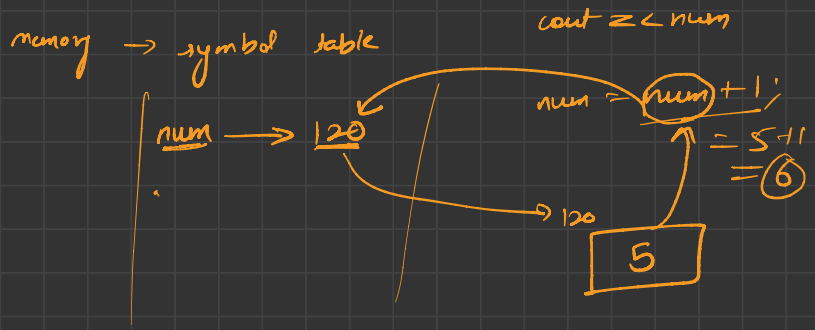
Pointers in C++





0x16b11f488 → Hexadecimal format

↳ 0 — 9 a b c def

pointer :- store addresses

```
int *ptr = &i;
```

create a pointer — address of i

int *ptr = &i

120

5

i

080

120

ptr

Bad Practice

int *ptr → pointer declare

{} → garbage value

ptr

int num = 5

int *p = &num

*p → Value at address pointed by p

5

100

p

100

5

num

int num = 5

100

5

num

int *p = &num

100

p

p

5

num

16

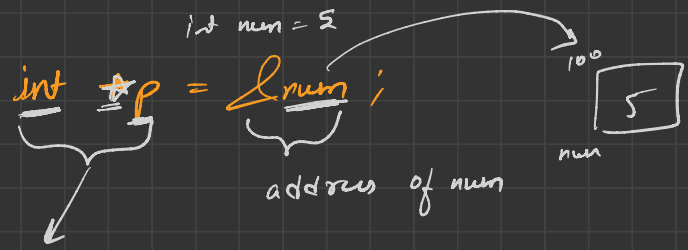
num = num + 1

(*p)++

p

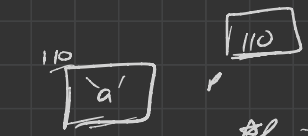
5

num



p is a pointer to int

char ch = 'a';



char *p = &ch

Access

int num = 5

↓

cout << num;

int *p = &num

cout << *p;

↓

dereference operator

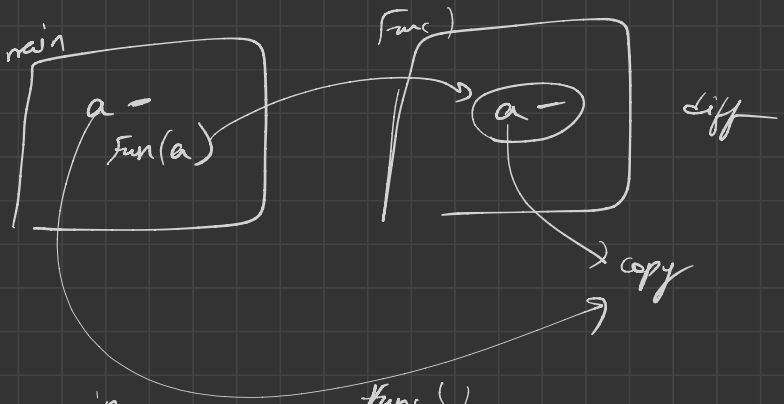
double d = 4.2;

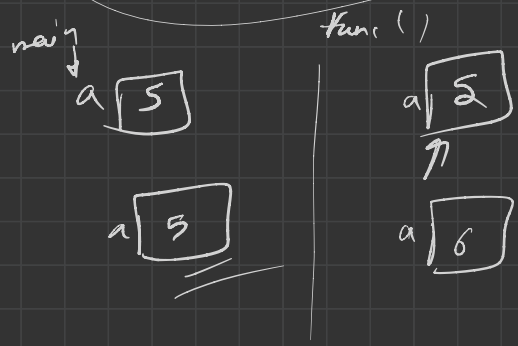
double *ptr = &d;

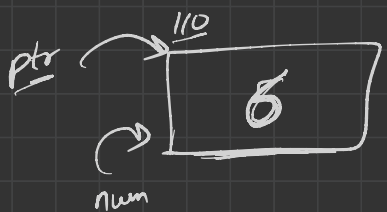
ptr is a pointer to double

*ptr = 4.2

num | *p → same value

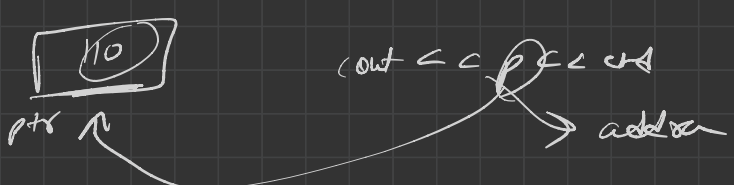


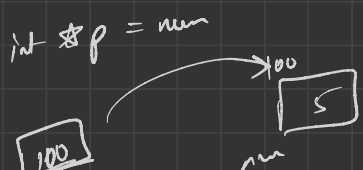




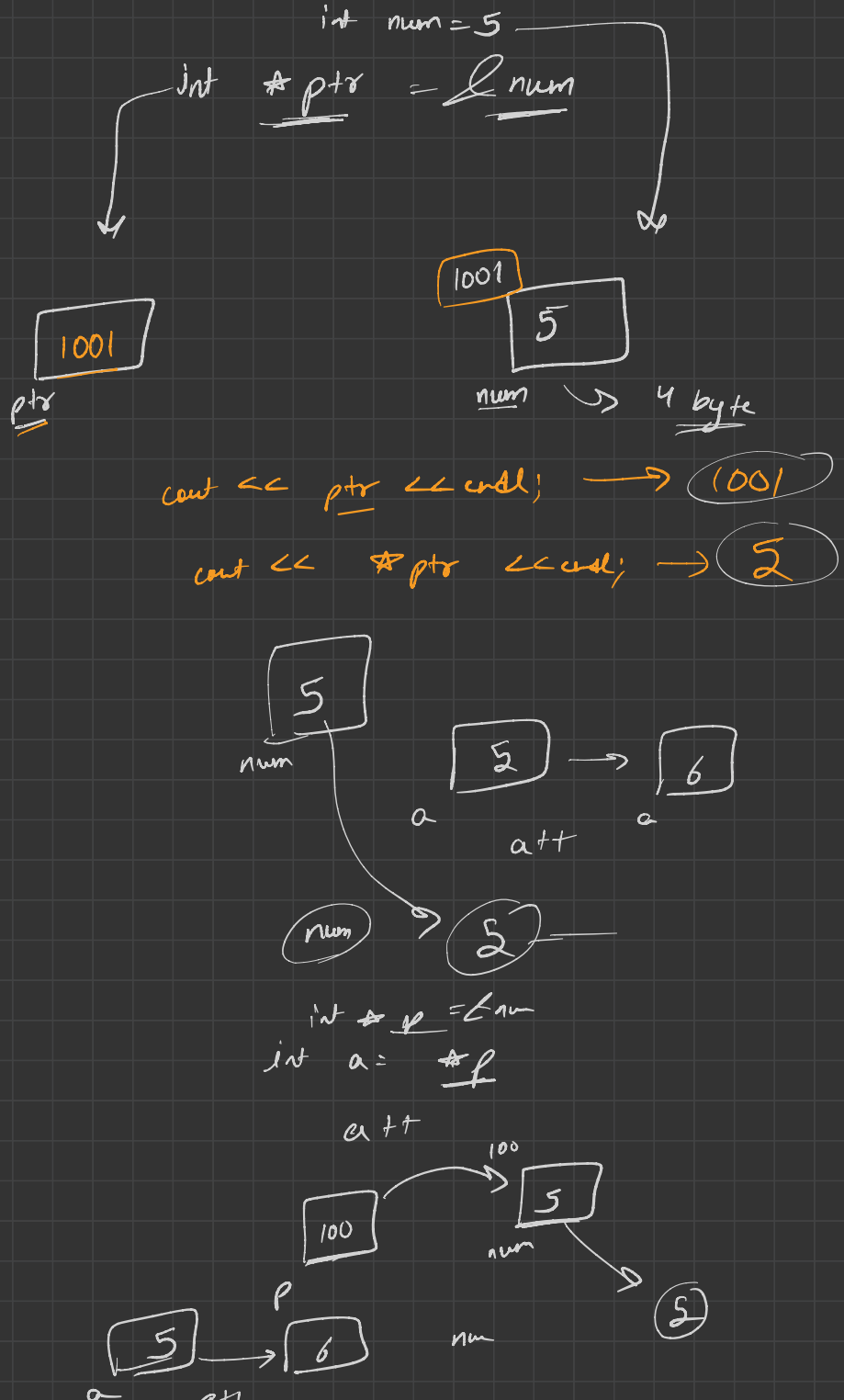
num ++

*ptr → 6

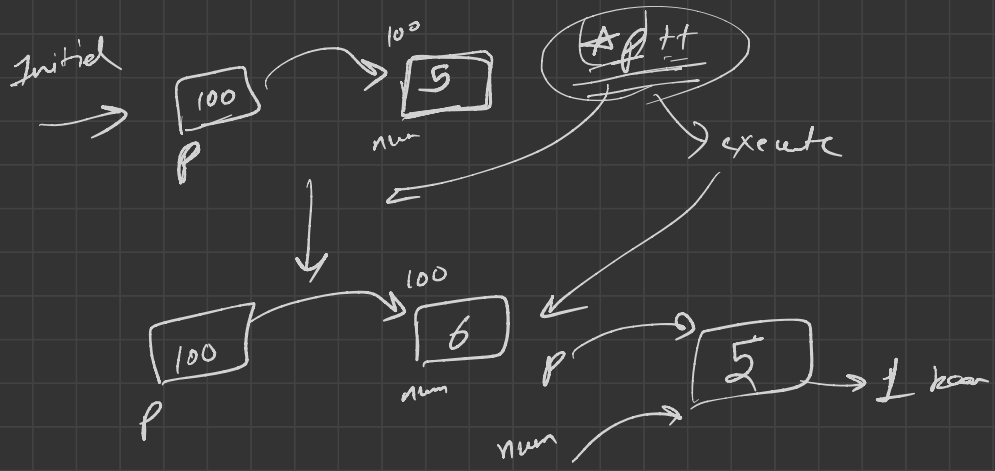




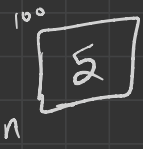
p (cout << p) → address of 6


int num = 5
int *ptr = &num
1001
ptr
1001
5
num
4 byte
cout << ptr << endl; → 1001
cout << *ptr << endl; → 5
5
num
5
6
a
a
a++
num
5
int *p = &num
int a = *p
a++
100
100
5
num
5
p
5
6
num
a
a++

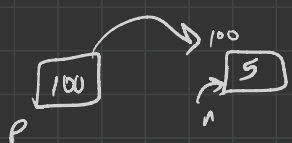
int *p = &num



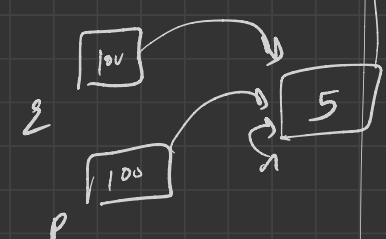
int n = 5



int *p = &n;



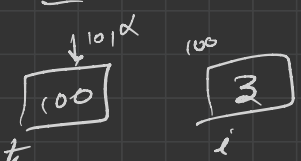
int *q = p;

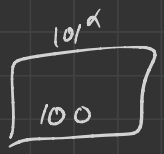


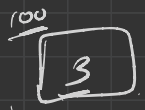
*t = *t + 1

Value at address stored in t = Value at address stores in t + 1

t = t + 1







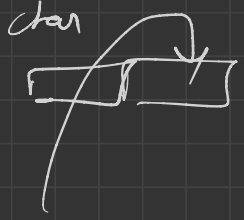
t

i

int -> 4

t = t+1;

104



100 -<104

int

char

t = t+1

100

108